

„Artificial Intelligence and Knowledge Engineering“ laboratory
summer semester 2025/2026
Version: 27th February 2026

Laboratory

- 1) Five exercises will be implemented during the semester.
 1. Problem solving by searching (deadline: 4th laboratory class).
 2. Game playing (deadline: 8th laboratory class).
 3. Symbolic Knowledge representation and its application in planning tasks (deadline: 9th laboratory class).
 4. Statistical Machine Learning (deadline: 12th laboratory class).
 5. Neural and Deep Learning (deadline: the last laboratory class).
- 2) Laboratory credit requires completing all exercises at least in a minimal form (see below).
- 3) Each exercise will be presented at ePortal on the web page for the lecture site in the separate files.
- 4) Each exercise will be evaluated from 0 to the maximum number points as specified in the description (Task 3 up to 50 pts, the other tasks up to 100 pts.).
In the case of exceptional or expanded fulfilment of the assignment the teacher may grant some additional score points.
- 5) Delayed exercises will result in a point penalty. Each delay of one laboratory class results in a 20% penalty in relation to the maximum.
- 6) Only one exercise may be presented and evaluated during one class.
- 7) Students MUST be prepared for each class in relation to the relevant exercise, both in terms of the exercise content and the theoretical foundations presented during the lecture. The instructor may verify student preparation by asking questions and may lower the grade for the exercise, if the student is not adequately prepared.
- 8) Up to two absences are acceptable. For each following absence the final grade is going to be lowered by 0.5.
- 9) If a student is engaged in something else during class, other than completing a laboratory assignment, they are deemed to be "inactive" in class, for which the instructor may deduct up to 20% of the maximum points for the current assignment (for each instance of "inactivity" identified).
- 10) The final grade is calculated on the basis of the percentage of the collected points according to the Study Regulations of the university.
- 11) An exercise is considered to be completed in a minimal form, if its evaluation is no less than 30% of the maximum.
- 12) It is possible to receive additional points for individual tasks, up to 20% of the maximum, if the student has demonstrated knowledge significantly beyond the scope of the course and has completed additional elements of the task previously agreed with the Teacher.
- 13) Cheating will be punished by the 0 points score. Repeated cheating will result in negative grade.
- 14) Students may use various available sources of knowledge (e.g., books or online materials), as well as sample programs (source code, if their license allows it!), but these MUST be properly cited in reports and in the source code submitted as a result of the exercise. A well-prepared report should include citations to the literature and other source materials. Any code fragments written by third parties MUST be appropriately identified by comments in the student's own program, and information about the original authors MUST be clearly identified in the headers

of the code files.

- 15) Similarly, Students may use artificial intelligence tools (AIT) when completing a task, but all cases of its use must be described in the task completion report and the elements of the solution (especially the source code) or report developed using NSI should be clearly marked.
- 16) Students are expected to possess very good knowledge about the whole presented solution including the whole submitted code.
- 17) The instructor may reduce the grade for an exercise based on the amount of foreign code or AIT-generated code used in the student's solution. If the amount of foreign code is significant, the grade for a given exercise should not exceed 70 points before late penalties are applied.

Lecture

- 1) Lectures are passed on the basis of the exam.
- 2) The dates of the two exam terms will be planned together with the faculty office and will be announced to students during the first half of semester.
- 3) Students may acquire additional scoring points during lectures on the basis of their activity (questions, answers to questions, remarks pertaining to lecture topics, etc.)
- 4) The collected scoring points during the semester will be a basis for increasing the result of the exam.

Course (*in the case of a group of courses*)

- 1) The whole course is passed, only if both the laboratory and lecture parts are passed.
- 2) The final grade is based on the arithmetic mean from the grades obtained for both parts.